

Stage 0 Recap: Node.js, Express & Git Fundamentals

1. Web Server Daemon & TCP Socket Binding

- **The Code:** `app.listen(3000)`
- **Sysadmin Analogy:** Think of this as launching a system service daemon (like `systemd` or `nginx`) that binds to a specific network socket on loopback (`127.0.0.1:3000`).
- **Route Handling (`app.get('/')`):** Acts like an inbound firewall or reverse-proxy routing rule. When an HTTP GET packet arrives at path `/`, Express processes the callback and returns an HTTP 200 OK header with the payload.

2. Process Management & Terminal Shells

- **Foreground Execution:** When you run `node server.js`, the Node runtime stays attached in the foreground to keep the listening socket open.
- **Multi-Tab Workflows:** To interact with a foreground daemon without killing it (`Ctrl + C`), you open a second terminal tab (a secondary SSH or local shell session) to execute client commands like `curl`.

3. Request/Response Inspection (`curl -i`)

- **The Command:** `curl -i http://localhost:3000/`
- **Why the `-i` flag matters:** It forces `curl` to display the raw HTTP response headers (HTTP/1.1 200 OK, Content-Type, Date, etc.) before the body payload. Inspection of raw headers is critical for API debugging and audit compliance.

4. Git Repository Scope & Local Identity

- **Scoped Configuration:** Running `git config user.email "your@email.com"` without the `--global` flag scopes your identity strictly to the local `.git/config` file in that project folder. Your global environment settings remain untouched.
- **Offline Ledger:** Git operates completely offline on your local machine. Committing saves snapshot hashes locally; connecting to a remote host (like GitHub) is a separate step coming in Stage 6.

5. Directory Hygiene & `.gitignore`

- **`node_modules/`:** Contains hundreds of third-party libraries. In systems management, committing `node_modules` is like backing up dynamic runtime packages or `/tmp/` files into a primary config git repo—it creates massive bloat.
- **`.gitignore`:** Tells Git to ignore build dependencies, keeping the repository clean, lightweight, and focused solely on your application source code.